

# The Neutrino Packages

dist · poly · finance · astro · rmt · phys · scatter — machine-verified

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## Contents

A package is a file of let definitions; `load("packages/name.nu")` runs it in the current session and its bindings persist. Records of closures act as namespaces (`norm.cdf`), and packages validate their inputs with `assert`, so they fail like builtins do — with a message, not an accident. Every transcript below is machine-verified against the interpreter by `tests/run_manual.sh`, and every numerical claim is cross-checked in the package's own golden tests against an independent reference (SciPy, NumPy, Python's `datetime`, or the `astral` library — named in each section).

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### 1. dist.nu — probability distributions

Six distributions — `norm`, `student`, `chi2`, `fdist`, `expo`, `unif` — each a record with `pdf`, `cdf`, `inv` (quantile), and `rand`. Parameters are explicit: `norm.cdf(x, mu, sigma)`. Quantiles with no closed form use bracket expansion plus `fzero` on the CDF; the quantile domain is  $0 < p < 1$ . Cross-checked against SciPy (`tests/34_dist.test`).

```
neutrino> load("packages/dist.nu")
neutrino> norm.cdf(1.96, 0, 1)
0.975002
neutrino> norm.inv(0.975, 0, 1)
1.95996
neutrino> student.inv(0.975, 30)
2.04227
neutrino> chi2.inv(0.95, 3)
7.81473
neutrino> fdist.cdf(3.0, 4, 20)
0.956799
```

A complete two-sample t-test — simulate, test, decide:

```
neutrino> load("packages/dist.nu")
neutrino> rng(42)
neutrino> let x1 = norm.rand(40, 100, 15)
[75.8016; 123.017; 111.725; 93.9971; 100.238; 98.0904; 107.158; 90.1486; 90.4082; 94.4609; 96.6851;
neutrino> let x2 = norm.rand(40, 108, 15)
[102.897; 92.0654; 112.122; 98.9768; 112.891; 112.931; 92.8931; 102.642; 111.903; 100.646; 100.94;
neutrino> let t = (mean(x2) - mean(x1)) / sqrt(var(x1)/40 + var(x2)/40)
2.46819
neutrino> 2 * (1 - student.cdf(abs(t), 78))
```

0.0157683

Out-of-domain probabilities refuse with the package's own message:

```
neutrino> load("packages/dist.nu")
error: quantile: p must be in (0,1), got 1.5
neutrino> student.inv(1.5, 10)
```

| Function                       | Worked example                  | Result   |
|--------------------------------|---------------------------------|----------|
| norm.pdf/cdf/inv(x, mu, sigma) | norm.cdf(1.96, 0, 1)            | 0.975002 |
| norm.rand(n, mu, sigma)        | mean(norm.rand(500, 10, 2)) > 9 | true     |
| student.pdf/cdf/inv(x, v)      | student.inv(0.975, 30)          | 2.04227  |
| chi2.pdf/cdf/inv(x, k)         | chi2.inv(0.95, 3)               | 7.81473  |
| fdist.pdf/cdf/inv(x, d1, d2)   | fdist.cdf(3.0, 4, 20)           | 0.956799 |
| expo.pdf/cdf/inv(x, rate)      | expo.inv(0.5, 2)                | 0.346574 |
| unif.pdf/cdf/inv(x, a, b)      | unif.cdf(3, 0, 10)              | 0.300000 |

## 2. poly.nu — polynomials

Coefficients are row vectors, highest power first:  $[2, -3, 1]$  is  $2x^2 - 3x + 1$ . Roots come from the companion matrix and eig — the same algorithm Octave uses, on Neutrino's LAPACK-verified eigensolver. Cross-checked against NumPy (tests/37\_poly.test).

```
neutrino> load("packages/poly.nu")
neutrino> polyval([2, -3, 1], 4)
21
neutrino> roots([1, -6, 11, -6])'
[1, 2, 3]
neutrino> roots([1, 0, 1])'
[1i, -1i]
neutrino> companion([1, -6, 11, -6])
[6, -11, 6; 1, 0, 0; 0, 1, 0]
```

Least-squares fitting (Vandermonde + backslash), and calculus that inverts:

```
neutrino> load("packages/poly.nu")
neutrino> let x = [0, 1, 2, 3, 4]
[0, 1, 2, 3, 4]
neutrino> let y = [1.1, 2.9, 9.2, 19.1, 32.8]
[1.1, 2.9, 9.2, 19.1, 32.8]
neutrino> polyfit(x, y, 2)
[1.95714, 0.131429, 1.01429]
neutrino> polyder([1, -6, 11, -6])
[3, -12, 11]
neutrino> polyint(polyder([1, -6, 11, -6]), -6)
[1, -6, 11, -6]
neutrino> conv([1, -1], [1, -2])
[1, -3, 2]
```

| Function         | Worked example                      | Result                                 |
|------------------|-------------------------------------|----------------------------------------|
| polyval(c, x)    | polyval([2, -3, 1], 4)              | 21                                     |
| roots(c)         | sort(real(roots([1, -6, 11, -6])))' | [1.00000, 2.00000, 3.00000]            |
| companion(c)     | companion([1, 0, -4])               | [0.00000, 4.00000; 1.00000, 0.00000]   |
| polyfit(x, y, n) | polyfit([1, 2, 3], [2, 5, 10], 2)   | [1.00000, -8.32667e-16, 1.00000]       |
| polyder(c)       | polyder([1, -6, 11, -6])            | [3, -12, 11]                           |
| polyint(c, k)    | polyint([3, -12, 11], -6)           | [1.00000, -6.00000, 11.0000, -6.00000] |
| conv(a, b)       | conv([1, -1], [1, -2])              | [1.00000, -3.00000, 2.00000]           |

### 3. finance.nu — the HP-12C's greatest hits

Time value of money, cash flows, bonds, amortization, and date arithmetic. Sign convention (HP-12C): money received is positive, money paid is negative; rates are per period, as decimals. Cross-checked against SciPy and Python's datetime (tests/38\_finance.test).

The mortgage block — payment, implied rate, savings growth:

```
neutrino> load("packages/finance.nu")
neutrino> pmt(360, 0.005, 250000, 0)
-1498.88
neutrino> rate(360, 250000, -1498.876313, 0)
0.005
neutrino> fv(120, 0.004, 0, -200)
30726.4
```

Cash flows and bonds:

```
neutrino> load("packages/finance.nu")
neutrino> npv(0.10, [-1000, 300, 420, 680])
130.729
neutrino> irr([-1000, 300, 420, 680])
0.163406
neutrino> bond_price(100, 0.08, 0.06, 10, 1)
114.72
neutrino> bond_ytm(114.7202, 100, 0.08, 10, 1)
0.06
neutrino> bond_mduration(100, 0.08, 0.06, 10, 1)
7.0236
```

amort returns the whole schedule as a record of columns — a small data frame you can index, sum, and plot:

```
neutrino> load("packages/finance.nu")
neutrino> let s = amort(1000, 0.01, 3)
{period = [1; 2; 3], payment = [340.022; 340.022; 340.022], interest = [10; 6.69978; 3.36656], principal = [12]}
neutrino> fields(s)'
["period", "payment", "interest", "principal", "balance"]
neutrino> s.payment[1]
```

```

340.022
neutrino> s.balance'
[669.978, 336.656, 4.26326e-12]
neutrino> sum(s.principal)
1000

```

Dates, HP-12C style — actual day counts, date arithmetic, day of week, and the bond market's 30/360 convention. `datestr` returns a display string, `daterec` the {y, m, d} record, and `today()` a serial day number, so arithmetic reads naturally — `datestr(today() + 90)` is the date in 90 days:

```

neutrino> load("packages/finance.nu")
neutrino> days(2026, 1, 1, 2026, 7, 9)
189
neutrino> datestr(datenum(2026, 7, 17) + 90)
"2026-10-15"
neutrino> daterec(datenum(2024, 2, 29))
{y = 2024, m = 2, d = 29}
neutrino> dateadd(2024, 2, 28, 2)
{y = 2024, m = 3, d = 1}
neutrino> dow(2026, 7, 9)
4
neutrino> days360(2024, 1, 31, 2024, 7, 31)
180
neutrino> daterec(today()).y >= 2026
true

```

| Function                                             | Worked example                                      | Result       |
|------------------------------------------------------|-----------------------------------------------------|--------------|
| <code>pmt(n, i, pv, fv)</code>                       | <code>pmt(360, 0.005, 250000, 0)</code>             | -1498.88     |
| <code>pv(n, i, pmt, fv)</code>                       | <code>pv(360, 0.005, -1498.88, 0)</code>            | 250001.      |
| <code>fv(n, i, pv, pmt)</code>                       | <code>fv(120, 0.004, 0, -200)</code>                | 30726.4      |
| <code>nper(i, pv, pmt, fv)</code>                    | <code>nper(0.005, 250000, -1498.88, 0)</code>       | 359.998      |
| <code>rate(n, pv, pmt, fv)</code>                    | <code>rate(360, 250000, -1498.876313, 0)</code>     | 0.00500000   |
| <code>npv(r, cfs)</code>                             | <code>npv(0.10, [-1000, 300, 420, 680])</code>      | 130.729      |
| <code>irr(cfs)</code>                                | <code>irr([-1000, 300, 420, 680])</code>            | 0.163406     |
| <code>bond_price(face, crate, y, n, freq)</code>     | <code>bond_price(100, 0.08, 0.06, 10, 1)</code>     | 114.720      |
| <code>bond_ytm(price, face, crate, n, freq)</code>   | <code>bond_ytm(114.7202, 100, 0.08, 10, 1)</code>   | 0.0600000    |
| <code>bond_duration(face, crate, y, n, freq)</code>  | <code>bond_duration(100, 0.08, 0.06, 10, 1)</code>  | 7.44502      |
| <code>bond_mduration(face, crate, y, n, freq)</code> | <code>bond_mduration(100, 0.08, 0.06, 10, 1)</code> | 7.02360      |
| <code>bond_convexity(face, crate, y, n, freq)</code> | <code>bond_convexity(100, 0.08, 0.06, 10, 1)</code> | 65.1716      |
| <code>amort(principal, i, n)</code>                  | <code>amort(1000, 0.01, 3).balance[3]</code>        | 4.26326e-12  |
| <code>datenum(y, m, d)</code>                        | <code>datenum(2026, 7, 17)</code>                   | 2.46124e+06  |
| <code>datestr(jdn)</code>                            | <code>datestr(datenum(2026, 7, 17) + 90)</code>     | "2026-10-15" |

| Function                                 | Worked example                                             | Result                                                |
|------------------------------------------|------------------------------------------------------------|-------------------------------------------------------|
| <code>daterec(jdn)</code>                | <code>daterec(datenum(2026, 7, 17))</code>                 | <code>{y = 2026.00, m = 7.00000, d = 17.00000}</code> |
| <code>dateadd(y, m, d, k)</code>         | <code>dateadd(2024, 12, 31, 1)</code>                      | <code>{y = 2025.00, m = 1.00000, d = 1.00000}</code>  |
| <code>today()</code>                     | <code>today() == datenum(now().y, now().m, now().d)</code> | <code>true</code>                                     |
| <code>dow(y, m, d)</code>                | <code>dow(2026, 7, 17)</code>                              | <code>5.00000</code>                                  |
| <code>days(y1,m1,d1, y2,m2,d2)</code>    | <code>days(2026, 1, 1, 2026, 7, 17)</code>                 | <code>197.000</code>                                  |
| <code>days360(y1,m1,d1, y2,m2,d2)</code> | <code>days360(2024, 1, 31, 2024, 7, 31)</code>             | <code>180</code>                                      |

## 4. astro.nu — the solar almanac

Sunrise, sunset, three grades of twilight, solar noon, day length, sun position, and moon phase, for any latitude and longitude (degrees; north and east positive). Times are local decimal hours given a UTC offset `tz`; `hm` renders them as “HH:MM”. NOAA solar position algorithm, within about a minute of the astral reference library (`tests/39_astro.test`). A `places` record preloads coordinates for several cities.

```
neutrino> load("packages/astro.nu")
neutrino> hm(sunrise(places.alexandria_va.lat, places.alexandria_va.lon, 2026, 7, 10, -4))
"05:52"
neutrino> hm(sunset(places.duluth_ga.lat, places.duluth_ga.lon, 2026, 7, 10, -4))
"20:50"
neutrino> hm(solar_noon(places.ljubljana.lon, 2026, 7, 10, 2))
"13:07"
neutrino> day_length(places.ljubljana.lat, places.ljubljana.lon, 2026, 7, 10)
15.5331
```

Planning a drive to arrive before dark — civil dawn at the origin to civil dusk at the destination:

```
neutrino> load("packages/astro.nu")
neutrino> let w = drive_daylight(places.alexandria_va, places.duluth_ga, 2026, 7, 10, -4)
{depart_dawn = 5.34382, depart_rise = 5.86728, arrive_set = 20.8376, arrive_dusk = 21.3161, window_h
neutrino> hm(w.depart_dawn) + " to " + hm(w.arrive_dusk)
"05:21 to 21:19"
neutrino> w.window_hours
15.9723
```

Sun position, moon illumination, and the honest polar refusal:

```
neutrino> load("packages/astro.nu")
error: the sun does not cross 90.833 degrees at latitude 80 on 2026-6-21
neutrino> let p = sun_position(38.8048, -77.0469, 2026, 7, 10, 13.22, -4)
{alt = 73.3594, az = 179.65}
neutrino> p.alt
73.3594
neutrino> moon_illum(2026, 7, 10)
0.19449
```

```
neutrino> sunrise(80, 0, 2026, 6, 21, 0)
```

| Function                                             | Worked example                                                                       | Result    |
|------------------------------------------------------|--------------------------------------------------------------------------------------|-----------|
| sunrise(lat, lon, y, m, d, tz)                       | hm(sunrise(38.8048, -77.0469, 2026, 7, 17, -4))                                      | "05:57"   |
| sunset(lat, lon, y, m, d, tz)                        | hm(sunset(38.8048, -77.0469, 2026, 7, 17, -4))                                       | "20:31"   |
| dawn_civil / dusk_civil(lat, lon, y, m, d, tz)       | hm(dawn_civil(38.8048, -77.0469, 2026, 7, 17, -4))                                   | "05:26"   |
| dawn_nautical / dusk_nautical(lat, lon, y, m, d, tz) | hm(dawn_nautical(38.8048, -77.0469, 2026, 7, 17, -4))                                | "04:48"   |
| dawn_astro / dusk_astro(lat, lon, y, m, d, tz)       | hm(dusk_astro(38.8048, -77.0469, 2026, 7, 17, -4))                                   | "22:23"   |
| solar_noon(lon, y, m, d, tz)                         | hm(solar_noon(-77.0469, 2026, 7, 17, -4))                                            | "13:14"   |
| day_length(lat, lon, y, m, d)                        | day_length(46.0569, 14.5058, 2026, 7, 17)                                            | 15.3443   |
| sun_position(lat, lon, y, m, d, hour, tz)            | sun_position(38.8048, -77.0469, 2026, 7, 17, 13.2, -4).alt                           | 72.2914   |
| moon_age(y, m, d)                                    | moon_age(2026, 7, 17)                                                                | 2.70686   |
| moon_illum(y, m, d)                                  | moon_illum(2026, 7, 17)                                                              | 0.0806581 |
| hm(h)                                                | hm(6.05)                                                                             | "06:03"   |
| places                                               | places.duluth_ga.lat                                                                 | 34.0029   |
| drive_daylight(from, to, y, m, d, tz)                | drive_daylight(places.alexandria_la, places.duluth_ga, 2026, 7, 17, -4).window_hours | 15.8284   |

## 5. rmt.nu — random matrices, structured

packages/rmt.nu is sugar over randn/qr/eye for the matrices you actually reach for at the prompt: symmetric, positive definite (chol-safe by construction: eigenvalues at least 1), Haar orthogonal, permutations and their matrices, correlation, row-stochastic (random Markov chains), and the Gaussian orthogonal ensemble — whose spectrum follows Wigner's semicircle on  $[-2, 2]$ , as the last line demonstrates on a  $200 \times 200$  draw. Everything is reproducible under `rng(seed)`.

```
neutrino> load("packages/rmt.nu")
neutrino> format(6)
neutrino> rng(11)
neutrino> randorth(2)
[0.684752, 0.728776; -0.728776, 0.684752]
neutrino> let P = randspd(3); chol(P)[1, 1] > 0
true
neutrino> randperm(6)
[4, 5, 1, 2, 3, 6]
neutrino> permuat([3, 1, 2])
```

```
[0, 0, 1; 1, 0, 0; 0, 1, 0]
neutrino> let H = goe(200); max(abs(eig(H).values))
1.99476
```

| Function     | Worked example                                                    | Result                            |
|--------------|-------------------------------------------------------------------|-----------------------------------|
| randsym(n)   | let S = randsym(3);<br>max(max(abs(S - S')))) == 0                | true                              |
| randspd(n)   | let P = randspd(3);<br>min(eig(P).values) >= 1                    | true                              |
| wishart(n)   | let W = wishart(3);<br>min(eig(W).values) > 0                     | true                              |
| randorth(n)  | let Q = randorth(3);<br>max(max(abs(Q' * Q -<br>eye(3)))) < 1e-12 | true                              |
| randperm(n)  | sort(randperm(5)) == 1:5                                          | [true, true, true, true,<br>true] |
| permmat(p)   | permmat([2, 3, 1])                                                | [0, 1, 0; 0, 0, 1; 1, 0, 0]       |
| randcorr(n)  | let C = randcorr(3);<br>max(abs(diag(C) - 1)) <<br>1e-12          | true                              |
| randstoch(n) | let T = randstoch(4);<br>max(abs(sum(T, 2) - 1)) <<br>1e-12       | true                              |
| goe(n)       | let H = goe(150);<br>max(abs(eig(H).values)) <<br>2.4             | true                              |

## 6. phys.nu — physical constants

packages/phys.nu is the CODATA 2018 constants as a record — exact where the 2019 SI redefinition makes them exact (c, h, k, NA, qe), the recommended values elsewhere. The transcript computes Earth's surface gravity from G, recovers the speed of light from Maxwell's relation  $1/\sqrt{\epsilon_0\mu_0}$ , and expresses thermal energy at room temperature in electron volts:

```
neutrino> load("packages/phys.nu")
neutrino> format(6)
neutrino> phys.c
299792458
neutrino> phys.hbar
1.05457e-34
neutrino> phys.G * 5.972e24 / 6.371e6 ^ 2
9.81997
neutrino> 1 / sqrt(phys.eps0 * phys.mu0)
2.99792e+08
neutrino> phys.k * 300 / phys.eV
0.0258520
```

| Function                | Worked example                           | Result      |
|-------------------------|------------------------------------------|-------------|
| phys.c                  | phys.c == 299792458                      | true        |
| phys.h                  | phys.h                                   | 6.62607e-34 |
| phys.hbar               | abs(phys.hbar * 2 * pi - phys.h) < 1e-45 | true        |
| phys.G                  | phys.G                                   | 6.67430e-11 |
| phys.k (thermal, in eV) | phys.k * 300 / phys.eV                   | 0.0258520   |

| Function          | Worked example           | Result      |
|-------------------|--------------------------|-------------|
| phys.NA           | phys.NA == 6.02214076e23 | true        |
| phys.R            | phys.R                   | 8.31446     |
| phys.qe           | phys.qe                  | 1.60218e-19 |
| phys.alpha        | 1 / phys.alpha           | 137.036     |
| phys.sigma        | phys.sigma               | 5.67037e-08 |
| phys.ly / phys.au | phys.ly / phys.au        | 63241.1     |

## 7. scatter.nu — scatter plots without touching the core

Proof that the frozen language plots more than it appears to: `plot()`'s `style = "points"` option is honored by every backend (SVG circles in the browser, point markers in `ascii` and `gnuplot`), so scatter plots are a pure package. `scatter(x, y)` and `scatter_titled(x, y, t)` wrap that fact; `jitter(x, amount)` spreads overplotted values for legibility. Per-point sizes and colors would need backend changes and are deliberately absent — the package's header says so. The transcript exercises the numeric helper; the SVG output itself is asserted by the plot test suite:

```
neutrino> load("packages/scatter.nu")
neutrino> rng(1); let j = jitter(1:100, 0.1);
neutrino> max(abs(j - (1:100))) <= 0.05
true
neutrino> size(jitter(rand(3, 4), 0.2))
[3, 4]
```

| Function                                                | Worked example                                                 | Result       |
|---------------------------------------------------------|----------------------------------------------------------------|--------------|
| <code>jitter(x, a)</code> stays within <code>a/2</code> | <code>max(abs(jitter(1:100, 0.1) - (1:100))) &lt;= 0.05</code> | true         |
| <code>jitter</code> preserves shape                     | <code>size(jitter(rand(3, 4), 0.2)) == [3, 4]</code>           | [true, true] |
| mean displacement is small                              | <code>abs(mean(jitter(zeros(1, 2000), 0.2))) &lt; 0.01</code>  | true         |

## Writing your own

The pattern all four follow: a file of `let` definitions, `assert`-validated inputs, a record of closures when a namespace helps, and a golden test file whose reference values come from an independent implementation. Multi-line definitions are fine wherever a bracket is open. `save` your workspace, `load` it tomorrow — a package is just a workspace you chose to keep.